

Arpan Pathak

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SUMMARY

Senior Software Engineer, **8 years**: developer tools, observability platforms, and performance-critical distributed systems at **Amazon, Microsoft, Oracle**. Full-stack **Python, JavaScript, Go, Rust, C++**; **Kubernetes, eBPF/Cilium**, GPU/CUDA, cloud **performance optimization**.

EXPERIENCE

Software Developer II Microsoft, Redmond, WA Jun 2025 – Present

- **Cluster Observability & Reliability**: Instrumented multi-region sovereign-cloud clusters with **Prometheus + Azure Monitor**, blue-green deployments, and automated validation, ensuring reliable releases across **AKS** clusters in France and Germany.
- **Kubernetes Operators & eBPF/Cilium**: Authored **Kubebuilder operators** reconciling network-policy state at scale; eBPF/Cilium-based L7 filtering and network visibility.
- **C++ → Rust Migration**: Rebuilt the Information Protection service from **legacy C++/C#** in **Rust**, moving memory safety to compile time; **~40% lower P99**, **2x throughput**, **3x lower memory/request**.

Senior Software Engineer Oracle Cloud Infrastructure, Seattle, WA Oct 2024 – Mar 2025

- **Developer-Facing SDK (Go)**: Built the **Terraform provider** for Oracle Autonomous Database, a developer SDK replacing legacy control-plane tooling; **\$1.2M/month projected savings**.
- **Database Engine Internals**: Optimized **lock-free data structures** in a proprietary DB engine, achieving **+45,000 ops/s** in high-concurrency read-write paths.

Software Development Engineer II Amazon, Seattle, WA & Hyderabad, India Mar 2021 – Oct 2024

- **Real-Time ML Inference (Python/PyTorch)**: **XGBoost + deep-learning ensemble** serving millions of ads/deals events at **100ms** across FreeVee, Twitch, Prime Video, driving **5% CTR lift** and **\$5.4M/month revenue growth**.
- **BERT NLP Service**: **5,000 QPS** at peak; cut support-ticket triage SLA from **7 days to 2 hours**.
- **Full-Stack Developer Platforms**: Led **React.js microfrontend** migration (deploy **5 days** → **30 min**); shipped an event-bus **SDK** (Kotlin, **<50ms** end-to-end) adopted by publisher teams.

SDE Razorpay (Go UPI gateway, 1M+ txns/hr, 99.99% uptime) & Mindfire (Java/Spring) 2018 – 2021

PERSONAL PROJECTS

Driving-CivicSense: Edge AI Vision System (Rust)

- Real-time on-device perception in **Rust** for lane & intersection awareness: **YOLOv8/v11** with pruning + quantization fitting tight memory/power budgets, **<50ms** inference on low-power ARM boards (Raspberry Pi Zero/5), fully on-device.
- Research paper: **Deterministic Intersection Blockage Prediction** | arpanpathak.github.io/driving-civicsense-vision-model/ | Code: github.com/arpanpathak/driving-civicsense-vision-model
- Books: **Seeing Machines** | arpanpathak.github.io/seeing-machines-book · **CUDA Kernels** (CUDA-Oxide / Rust-to-CUDA) | arpanpathak.github.io/gpu-parallel-book

SKILLS

Languages: Python, JavaScript/TypeScript, Go, Rust, C++, Java, Kotlin

Systems & Infra: Kubernetes, Kubebuilder, Docker, microservices, eBPF/Cilium, Linux, computer architecture

Observability & Profiling: Prometheus, Grafana, Azure Monitor, CloudWatch, trace/sampling, performance profiling

GPU & Performance: CUDA, GPU kernels, P99/latency tuning | **Web & APIs**: REST, gRPC, Protobuf, React.js

EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018